

Forecasting financial signal for automated trading: An interpretable approach

Mohammad Reza Roostaei, Ahmad Ali Abin*

Faculty of Computer Science and Engineering, Shahid Beheshti University, Tehran, Iran

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ABSTRACT

Traders can now use online trading applications to increase their capital by benefiting from financial markets fluctuations. The foreign exchange (forex) market, the silver and gold exchange markets, and stock indices (e.g. S&P 500) are among the most famous financial markets. Traders must identify the correct trends in any given market and take appropriate actions to increase their earnings. High levels of knowledge and competency are required to conduct profitable transactions. Deep learning and data clustering were employed in this study to propose an interpretable automated financial market trading model. In the proposed method, feature vectors are first extracted from the market price index and the values of useful indicators (e.g., RSI and MA). Data labeling is then performed through clustering. After that, a dataset is created to train a deep learning model in order to identify and adopt the best action (buy, sell, or no action). In financial markets, prices may change drastically and disrupt the performance of learning models. Therefore, an autoencoder network was designed to identify drastic fluctuations (outliers) in the market and avoid trading at such cases. This could prevent a steep drawdown. Experiments were performed on different markets to compare the proposed approach with the state-of-the-arts. The results indicate higher profitability with respect to the trading risk in comparison with other methods.

1. Introduction

The financial marketing industry is globally known to include markets with high trading volumes. Operating in various areas, these markets allow individual traders and trading companies to participate in trade. The financial markets of gold, silver, stocks, and foreign exchange in many countries are known as stock exchange markets worldwide. These trading markets have an enormous potential to increase earnings. It is possible to make considerable investments in these markets. However, traders (investors) must have high technical knowledge about trading to identify the best time for trading and take the best action in order to prevent loss and reduction of the initial capital. Traders must constantly monitor the trend in price fluctuations to find the best time for trading. This is a very time-consuming task. For trading in such markets, traders use various techniques, one of which is to employ the frequent patterns in the market price trend. Usually, technical patterns emerge in market price trends. The right decision can be made for future trade by detecting these patterns; however, it is not easy for beginners to identify these patterns. Nevertheless, machine learning techniques can be employed to detect these patterns. Hence, machine learning methods and automated training can be regarded as appropriate strategies in these markets.

Machine learning approaches are known as powerful methods for time series forecasting, i.e. consecutive datasets that change over time. Additionally, the earlier study showed that ANN methods are more accurate in nonlinear combination forecasts than individual forecasting methods (ARIMA, Brown's exponential smoothing method, and trend analysis) and linear combining (equal weights and minimum-variance) methods (Shi et al., 1996). For instance, consider the studies on forecasting time series data in meteorology (Vasanth et al., 2019), cryptocurrencies (McNally et al., 2018), and computer networks (Chen et al., 2017). In some cases, forecasting time series could influence the plans and strategies of massive organizations. For example, prediction of electricity consumption by an accurate approach like DeepESN and comparing that with anticipated energy generation can inform the necessary changes and developments in the electricity generation industry (Hu et al., 2020). Since all financial market data are basically considered time series data, they can be employed for price forecasting in financial markets through existing solutions in different areas (Xu et al., 2020). As a result, it will be possible to adopt the best action such as buy, sell, or no action.

* Corresponding author.

E-mail addresses: m.roostaei@mail.sbu.ac.ir (M.R. Roostaei), a_abin@sbu.ac.ir (A.A. Abin).

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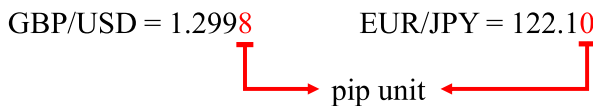


Fig. 1. The pip unit for two forex symbols.

There are usually two main challenges in forecasting and adopting the right action in financial markets. First, market prices change drastically beyond expectations in a specific interval. In this case, the data from that interval are called anomaly. Anomalies in the forecasting process constitute a serious challenge because drastic fluctuations in the market price trend can decrease the decision accuracy of the model trained to adopt the best action.

In forecasting economic signals and taking the right action, the next challenge emerges when the decision-making model requires labeled data in its learning process. However, the existing data are the market's raw signals with no labels to determine which action was the best choice for each time t of the market. The next important challenge is how to select the right label for data. This paper aims to select the best action¹ (buy, sell, or no action) in a smart way. In other words, the research problem is converted into a classification problem by selecting the most appropriate label for data. An efficient and optimal labeling method must be adopted due to the importance of labels selected in the network training process and their impacts on the accuracy and performance of the decision-making system.

This paper proposes an interpretable model to take the best action in financial markets. The two above-mentioned challenges are taken into account to solve the research problem through an appropriate approach. A deep learning model was used in the proposed approach for making decisions on the best action. Clustering and optimal labeling techniques were employed to create and label the training dataset. Furthermore, the idea of anomaly detection was used along with autoencoder networks to detect drastic fluctuations (outliers) in market trends.

1.1. A review of financial markets

There are specific terminologies used in the area of financial and stock markets, some of which are used in this study too. The following section introduces the forex market in general and discusses some relevant terminologies.

In financial markets, there are different sectors such as the securities exchange as well as gold and silver markets for investment and trading. A popular sector is the foreign exchange, abbreviated to forex. In these markets, every ticker symbol includes two currency units (e.g. EUR/USD), and its price is calculated by considering the value of the first currency in proportion to the second currency. For instance, if the price of EUR/USD is 1.3, it means that each euro is worth 1.3 US dollars. In the forex market, the price unit is called pip, which stands for point in percentage. In fact, one pip is a ten-thousandth of the price of a pair of currencies. However, when the Japanese yen (JPY) is one side of a pair of currencies, every pip is worth one-hundredth of the price. Fig. 1 shows the place value of each pip in the price for two ticker symbols.

In financial markets, prices increase or decrease. An advantage of forex markets is profitability when there are positive and negative fluctuations in prices. In other words, a trader opens a buying order by forecasting a rising price change. It is also possible to open a selling order by forecasting a falling price change. When the order reaches the expected profit threshold, the trader closes the order and withdraws

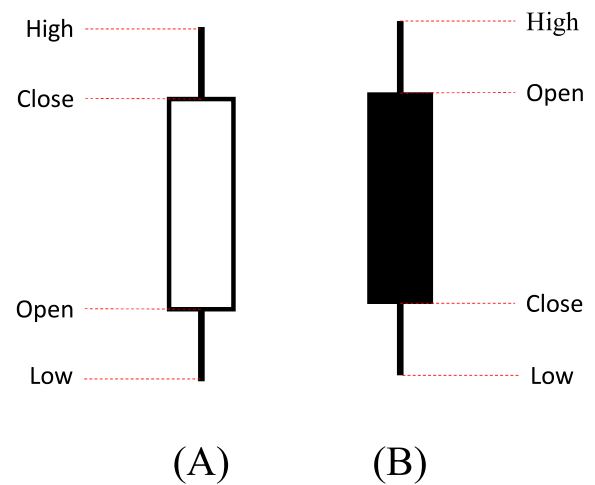


Fig. 2. Two different candlesticks: (A) rising candlestick and (B) falling candlestick.

the profit. In simple terms, opening a buying order for the X/Y symbol means purchasing the value ratio of X to Y, and the profit or loss of the trade is received when the order is closed. In a selling order for this symbol, the trader purchases the value ratio of Y to X and receives the profit when the order is closed.

Different charts are used to display the financial market data. Candlestick chart is one of the most useful diagrams. In this chart, a candlestick is created for each time frame. If the opening price is lower than the closing price, the candlestick is rising; otherwise, it is a falling candlestick. Fig. 2 shows two rising and falling candlesticks.

For profitable trading, it is necessary to accurately identify the trade type (selling or buying) and the right time for opening and closing orders. For this purpose, the retrospective indicators and candlestick patterns of the market are used. Fig. 3 shows the EUR/USD price trend. It presents a market segment in which a drastic price fluctuation was recorded. This type of price data pattern is called an outlier pattern.

1.2. Literature review

Trading in different capital and securities markets can be very appealing to many individuals who want to increase their capital. Therefore, many machine learning researchers have tried to facilitate forecasting in these markets. There is a diversity of approaches that tried to forecast and trade automatically. For instance, some of them, like the high-frequency trading method, need software and hardware implementation because of the importance of connection speed (Huang et al., 2019). This section reviews the relevant papers and approaches. These papers have been categorized by using their main model of approach in Table 1.

Selecting a suitable feature set for forecasting the changes in financial performance has been a challenging problem. However, Jiang et al. (2009) showed that the genetic algorithm could select the appropriated influencer features on financial performance. According to their study, it can be seen that after training the model, each of the considered industries, which included China's listed companies, had a different feature selected set.

Galeshchuk and Mukherjee (2017) employed a CNN to develop a model for forecasting the direction of financial markets. Their proposed approach outperformed basic methods such as the SVM. Their datasets included EUR/USD, GBP/USD, and JPY/USD symbols.

Singh and Srivastava (2017) proposed a method with a general 2-directional and 2-dimensional (PCA) structure along with a deep network. They compared their proposed method with the 2D PCA+RBFNN (radial basis function neural network) and showed its superiority. For evaluation, they used a stocks dataset from Google Co. on NASDAQ.

¹ In this study, the meaning of buy (sell) action is opening the long (short) position in financial market.

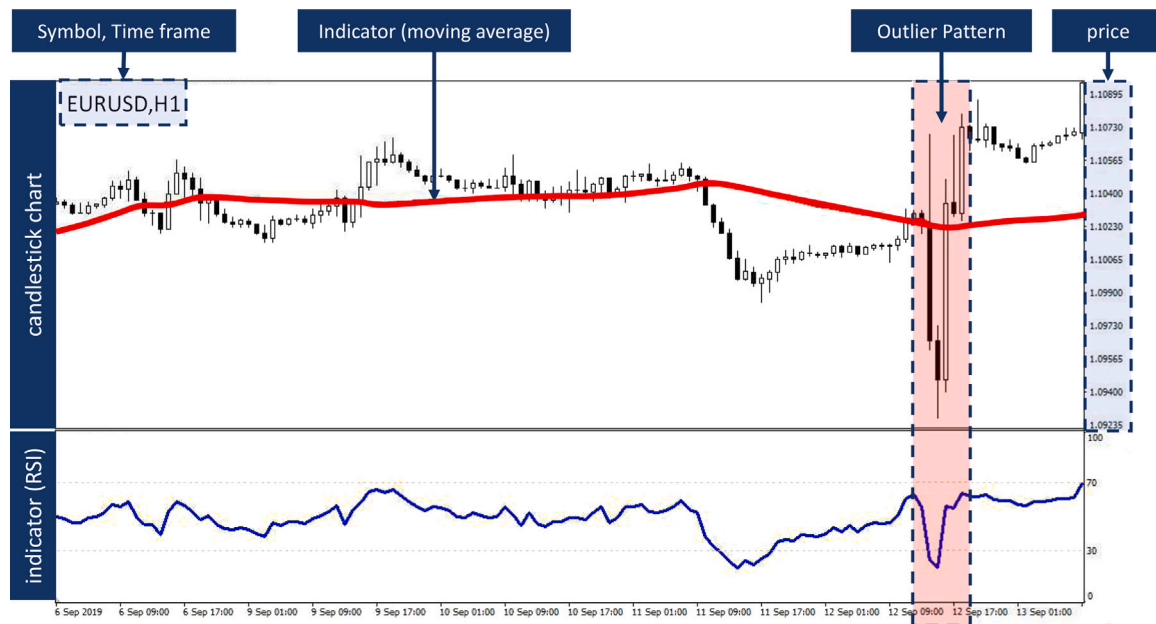


Fig. 3. An illustration of the market segment through two indicators and a case of market outlier.

Table 1
Some state-of-the-arts in financial market forecasting along with their core idea.

Core idea	Ref.
Genetic algorithm	Jiang et al. (2009)
Deep network	Galeshchuk and Mukherjee (2017), Hosaka (2019), Singh and Srivastava (2017) and Yan and Ouyang (2018)
Deep belief network	Shen et al. (2015) and Zheng et al. (2019)
Deep Network+Gradient-Boosted Tree+Random Forest	Krauss et al. (2017)
Chaos theory+Multi-layer Perceptron+Multi-Objective	Ravi et al. (2017)
Evolutionary Computing	
Markov chains	Wilinski (2019)
Fuzzy logic	Chen and Chen (2015)
Elastic network	Contreras et al. (2018)
Deep direct reinforcement learning	Deng et al. (2017)
CNN-LSTM	Lee and Kim (2020) and Rezaei et al. (2021)
LSTM	Shen et al. (2021)
CNN-LSTM (news-based)	Cai et al. (2018) and Lee and Soo (2017)
Decision tree (news-based)	Carta et al. (2021)

Yan and Ouyang (2018) introduced a model based on LSTM for long-term prediction of prices using Shanghai Composite Index dataset in a daily timeframe. They also proposed the wavelet decomposition and reconstruction of financial time series to increase the long-term forecasting accuracy. They compared the forecasting ability of their proposed approach with such methods as SVM, MLP, and K-NN. The results showed the superiority of their proposed method.

Hosaka (2019) used imaged financial statement to predict bankruptcy. They used imaged financial statements obtained from 102 bankrupt and 2062 active companies listed on the Tokyo Stock Exchange. In the first phase, the images were converted into grayscale versions. They were then used to train a CNN-based model. Tadaaki's proposed method outperformed the techniques that rely on generalized and statistical data. Comparing the proposed method with basic techniques such as decision trees, MLPs, SVMs, and AdaBoost, Tadaaki managed to predict bankruptcy likelihood more accurately.

Shen et al. (2015) used a deep belief network (DBN) to forecast fluctuation rates. They employed continuous restricted Boltzmann machines (CRBMs) to train a DBN. For this purpose, they used the conjugate gradient method to accelerate the DBN training phase. This method outperformed other typical techniques such as the feedforward neural network (FFNN).

Zheng et al. (2019) employed a DBN, defined network layers and parameters, and obtained their optimal values to help the trained

network to outperform the FFNN. Their study focused on INR/USD and CNY/USD data. According to the research results, their proposed approach outperformed the FFNN in these symbols.

Krauss et al. (2017) used three models in an ensemble including deep neural networks, gradient-boosted trees and random forests, which were equal-weighted to predict one-day-ahead stock returns for the S&P500. Their proposed approach was considered a statistical arbitrage, i.e. a strategy for short-term financial equations based on mean revision analyses. According to the results reported from October 1992 to December 2015, the maximum earnings of this hybrid model was 0.45% on a daily basis with 0.73% of max drawdown.

Ravi et al. (2017) proposed a method for financial time series prediction in three stages named the chaos (detecting anomalies through the chaos theory), a multi-layer perceptron (or an MLP for adopting the best action), and the non-dominated sorting genetic algorithm (or the NSGA-II for finding the optimal parameters). The datasets used by Ravi et al. included JPY/USD, GBP/USD, and EUR/USD symbols.

The main idea of Wilinski (2019) is separating time series of signals in the stock market into the fixed-size window and consider them as states of Markov. The aim of optimization of this Markov model consist to find the best length for windows, the number of windows, and the number of intervals. Symbols which they worked on are EUR/USD in the hourly time frame and WIG20 – the most Polish stock exchange index – in the daily time frame.

Chen and Chen (2015) used the second-order fuzzy relationship and the probabilities of trends to forecast trend of market. In the first step, they formed the second-order fuzzy logical relationships of the main factor and then they fuzzified the historical training data of both the main and the secondary factors into fuzzy sets. Afterwards, they grouped the obtained second-order fuzzy relationships into two-factors second-order fuzzy-trend logical relationship groups. Finally, they calculated the probability of mapping each trend to one of “down-trend”, “equal-trend” and “up-trend” categories. They evaluated their proposed approach to Taiwan Stock Exchange Capitalization Weighted Stock Index (TAIEX), NTD/USD, JPY/USD, CNY/USD and KRW/USD symbols.

Contreras et al. (2018) analyzed the interplay between currency pairs on the forex market to assume the currency symbols in an elastic network which always tries to maintain stability. With this argument, they predicted the market price trend in a pair of currency symbols by using the interplay of changes in the value of symbols. In their proposed elastic networks, the nodes included 7 currencies used in the forex symbols, whereas the edges show the effects of currencies on each other (the greater the mutual effectiveness of the two currencies, the shorter the edge between their corresponding nodes).

Deng et al. (2017) employed a fuzzy deep direct reinforcement (FDDR) method to adopt the best action (sell, buy, or no action) instantly. In the fuzzy block of this method, the market price trend is connected to three rising, falling, and constant nodes for every input node. The fuzzy block output is then sent to the deep layer block in a flat manner. The retrospective outputs of the model are injected as the inputs into the hidden layers of the deep network to use the effects of the previous actions in making the right decision for the moment. This deep learning network was trained by using the earning as the reward function. The dataset used in this paper included data of IF, AG, and SU.

Lee and Kim (2020) proposed their model for predicting the close-price value of the next day in markets. Their strategy consists of two main feature extractors (super-high dimensional market information feature extractor and target index feature extractor) which are compound together for prediction close-price value. They called this model NuNet. They assessed Nunet as an Autotrader with the condition-based equation; If the difference of predicted value and real previous close-value has enough distance, the system will consider suitable action (by rising-prediction, buying action will be considered, and by falling-prediction, selling action will consider). The Nunet has been assessed in S&P500, KOSPI200, and FTSE100 and has had 69.32%, 61.02%, and 22.84% profitability in investigated markets respectively.

Rezaei et al. (2021) structured a deep learning model to predict the value of the market of the next day by decomposition of frequency of market signals. In the first step of their model, the signal will be decomposed. Then, decomposed signals will be injected into separation CNN-LSTM branches. At the last part of these branches, values of branches will be added together and represent the close price of the next day. Stock markets that have been assessed in that paper are S&P500, DowJones, DAX, and Nikkei225.

Shen et al. (2021) claimed that by considering an economic theory, the two-way causal relationships present in trade data, the future values forecasting would be more accurate. So, they proposed an LSTM model with 2-dimension input (multivariate time series). The first one is import and the second one is the export amount of a country. They trained and evaluated their approach in ten countries in the monthly time frame (M1). The study metrics were RMSE and MAPE for each export and import output. They used a classical approach, ARIMA, as their baseline. Then, they showed that their model improved RMSE and MAPE on the export forecasting by 17.048% and 1.463%, respectively. Also, they improved RMSE by 40.939% and MAPE by 1.806% on the import forecasting.

Although prediction of financial markets depends on some technical approaches, like using indicators on the price fluctuations, news and

fundamental happening can directly influence the following prices in markets. Cai et al. (2018) proposed an ensemble method for a classification problem (falling or rising trend). Their ensemble method consists of 7 models CNN and LSTM based. The CNN models focused on news titles, news keywords, news content, and the LSTM models tried to predict falling and rising by forecasting the following second close price. They also quantized texts (news) and introduced a tendency dictionary for determining the tendency of the market to fall or rise after publishing news.

Lee and Soo (2017) proposed a recurrent convolutional network (RCN) to predict the price of financial markets. The main idea of their study was to convert news titles to vectors by a word2vec CNN-based and use these vectors and historical prices as an input vector of an LSTM block to predict the next price. Furthermore, they divided the data set of each considered stock into three categories (rise, fall, and neutral) and trained the model as a classifier, which is called RCN-C. They trained and evaluated their model from 2007 June 21st to 2017 Feb 13th in five companies on Taiwan Stock Exchange (TWSE). They showed that the combination of their proposed model with some technical algorithms like Stochastic Oscillator (KD) and Moving Average (MA) could be more profitable than purely technical approaches in some stocks.

Some news has a direct impact on the price of stock markets. Carta et al. (2021) proposed a model that can predict the next day variation (without concentration on the positive or negative difference) in individual companies of S&P500. In their study, the lexicon of news is generated and categorized in some distinguished industries. In the next step, for the high-low price variation prediction on the following day of each company, they considered the company-related and industry-related news in short-, mid-, and long-term (previous day, week, and month) to generate the feature vectors. Then, the prepared vectors are fed to a decision tree classifier to predict the high or low variation of the next price. They used a walk-forward approach for validation from 2004 to 2016 and tested their method from 2016 to the end of 2017.

1.3. Contributions of this paper

This study proposed a system for automated trading through deep learning. An important issue in using deep learning for forecasting is the interpretability that is the degree to which a human can understand the cause of a decision. However, most of the existing works in forecasting financial signals suffer from poor or no interpretability of the models. In this study, we used the idea of data clustering along with deep learning to give interpretability to the output of the proposed model. Another innovation of the proposed method is to use an autoencoder network for the detection of outliers to prevent the effects of drastic fluctuations on the performance of the decision-making model for taking the best action. This paper also used the values of useful indicators in addition to the market price as the most important feature in the decision-making model input. Moreover, the system performance evaluation criteria were selected by considering the proposed system as an automated trading machine. Many studies on financial market prediction focus on predicting the average market price for the next temporal unit. However, this parameter is not sufficient for trading, and it is necessary to consider other parameters such as profit and loss limits and the drawdown rate. Accordingly, total profit and maximum drawdown were used in this study to evaluate the performance of the proposed method.

1.4. Organization of this paper

The paper is structured as follows. Section 2 presents the proposed method. Section 3 addresses the implementation of the proposed method along with the parameters setting and the dataset. In Section 4, the results are presented and analyzed. Finally, Section 5 draws a conclusion.

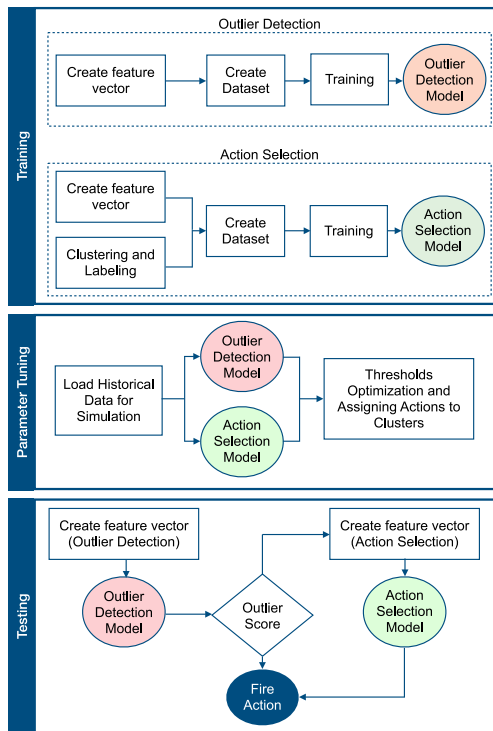


Fig. 4. An illustration of the proposed approach in different phases.

2. Proposed method

As mentioned in Section 1, the financial market data are basically time series data which might follow unpredictable and drastic patterns for various reasons such as the news, war, and peace treaties. This paper uses a method based on outlier detection to first discover drastic fluctuations in the market. After that, the best action (sell, buy, or no action) is adopted, and transactions are finally conducted in an automated manner. The idea of the proposed method consists of two separate models which are placed consecutively. These two models were trained in separate training phases. The first model aims to detect anomalies or severe fluctuations in the market, whereas the second model is responsible for adopting the best action at the moment. The second model adopts an action when the outlier detection model sees no fluctuations in the market. If the market is identified as volatile, the conservative action is no action. In addition, extracting the high-level features is one of the noticeable abilities of deep learning approaches. Because of the size of input vectors in both models, which had to be trained as outlier detection and action selection units, and complex relevances between features of input vectors, we decided to use deep learning models. Fig. 4 shows the general process of each phases (training, parameter tuning, and testing) of the proposed method. Also, Fig. 5 illustrates the blocks of testing phase. For clarification, the pseudo code for the proposed method is presented in Algorithm 1. Details for each block is given in the following sections.

2.1. Outlier detection unit

There are usually some anomalies and outliers in the training dataset. Due to the low frequency of outliers in the training process, machine learning models fail to be trained sufficiently in how to react to these types of data. The financial market data is no exception. Therefore, outliers are eliminated from the action process in order to increase accuracy in making the best decisions. Although a significant number of anomaly detection methods exist in the literature (Li et al.,

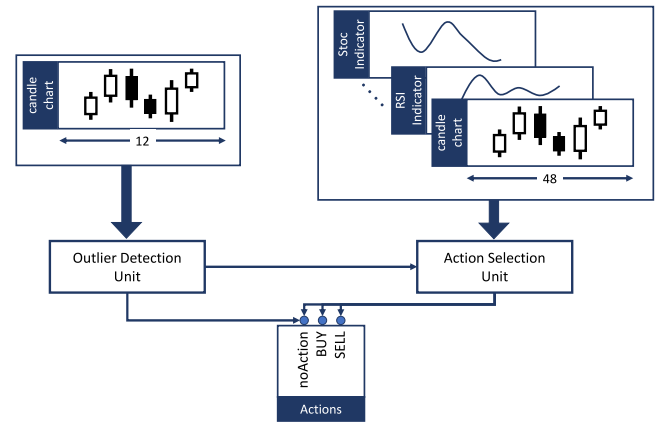


Fig. 5. An illustration of the proposed forecasting model in testing phase.

Algorithm 1 The proposed method

- 1: Create training set (set of vectors and cluster id's as label)
- 2: Train the outlier detection model
- 3: Train the action selection model
- 4: Find the optimal threshold for outlier detection
- 5: Select the best action for each cluster
- 6: Create test vector for the current state of market
- 7: Check test vector for outlier by using the outlier detection model
- 8: **if** pattern is outlier **then**
- 9: **return** noAction
- 10: **else**
- 11: **return** best action by using the action selection model
- 12: **end if**

2020, 2018). In this study, the autoencoder networks are employed to train a model, the outputs of which are then used to determine an outlier score for specific data. Comparing the resulting anomaly score and the anomaly threshold obtained from the optimization phase, it is possible to determine whether the input data are identified as outliers. This unit is introduced in detail in the following section.

2.1.1. Using autoencoder network for outlier detection

Fig. 6 shows the general structure of the outlier detection unit. This type of convolutional neural network is called an autoencoder network which consists of two main elements: an encoder and a decoder. These networks are used to recreate the input vector in the output. In other words, after the training phase and encountering training datasets, the network can create the input vector in the output layer by considering similar features observed in the training data. With this argument in mind, if the components of the input vector have not been encountered by this unit in the training process, the output vector will be significantly different from the input one. The more similar the output vector to the input vector, the lower the level of anomaly in the data sequence. In this study, for each one-hour timeframe t , a 12-dimensional feature vector is created by using the average price value of its 12 previous time frames and used to train the autoencoder. Fig. 7 shows two output vector samples of the outlier detection model.

2.1.2. Calculating anomaly score

Feeding the outlier detection unit by an input sequence will return a sequence as the output vector. The similarity of which to the input vector can help determine the anomaly score of the input sequence. A weighted comparison is drawn between the input and output vectors of the outlier detection unit to determine the level of anomaly in the anomaly score measurement unit.

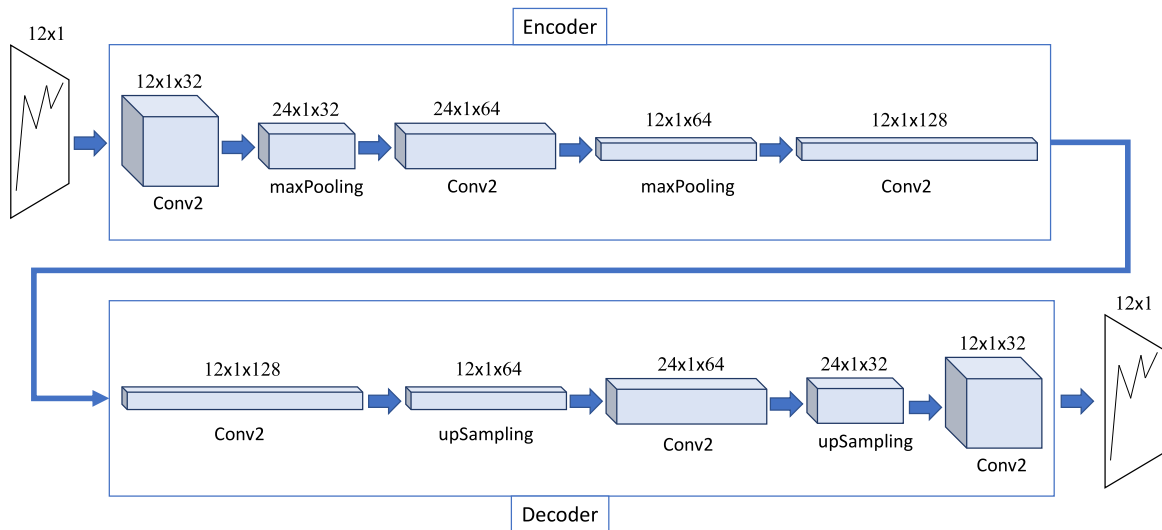


Fig. 6. The proposed autoencoder network for outlier detection.

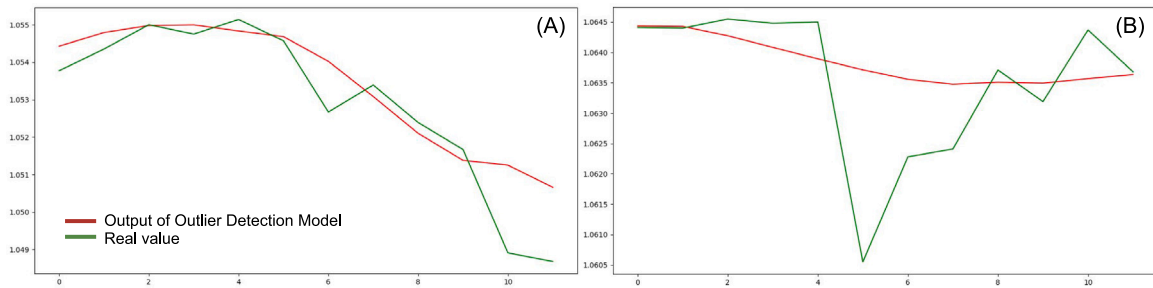


Fig. 7. (A) Output vector of the outlier detection model and real values of a normal data and (B) an outlier data.

According to Eq. (1), the components of input and output vectors are compared. In fact, this equation can detect anomaly sooner if the components of the input vector are more different from the corresponding components in the output vector. Therefore, the chances are higher for conducting loss-making transactions by considering high-frequency patterns in the market retrospectively.

$$Score_{outlier} = \sum_{i=1}^{12} i \times |X_i - Y_i| \quad (1)$$

In Eq. (1), X_i shows the i th component of the input vector, whereas Y_i indicates the i th component of the output vector in the autoencoder network model for outlier detection.

Outliers can be detected by adjusting the anomaly threshold in proportion to the symbol on which a transaction is going to be conducted. Then the trading process is stopped for the outliers. The best threshold is selected in the optimization phase (Section 3.3).

2.2. Developing action selection model

Once it is made certain that there are no anomalies in the data, it is necessary to adopt the best reaction. In this unit, a deep model was employed to classify data as sell, buy, or no action. The following steps were taken to develop this model.

Step 1: Creating feature vectors

First, the feature vector is extracted for each moment t of the market. An appropriate label is then selected and assigned to each feature vector. These vectors are then used as the network input. Based on their labels, the network is trained and evaluated. The following section explains how to create and label feature vectors.

Table 2

Features used in the input vector of the decision-making network.

#	Feature	Description
1	avg_price_value	Average market opening and closing price
2	volume	The number of transactions in a candle
3	close_value	The market price at the closing moment of a candle
4	open_value	The market price at the opening moment of a candle
5	low_value	The minimum market price along the candle
6	high_value	The maximum market price along the candle
7	stoc_stoc_value	The first output of the stochastic oscillator
8	stoc_signal_value	The second output of the stochastic oscillator
9	ma_value_20	The output of the moving average within an interval of 20
10	ma_value_50	The output of the moving average within an interval of 50
11	rsi_value	The output of RSI (Relative Strength Index)

■ **Creating feature vector for each time t :** In financial markets, it is possible to take advantage of useful indicators in addition to the instantaneous price at time t . Also, the earlier study showed that trading volume can significantly influence financial forecasting (Zhu et al., 2008). Different indicators were used in the proposed approach to create the feature vector. In this study, 11 features for each 48 past candlesticks ($t_{-47}, \dots, t_{-1}, t_0$) of the market at time t were extracted as a 2D feature vector. That means for each time t , a matrix with the shape of 11×48 is created as the feature vector. Table 2 shows these 11 features extracted for each time $t_{i:i=-47,\dots,0}$ of the vector.

■ **Assigning label to feature vector:** A clustering approach was used to assign label to each extracted feature vector. Here, the idea is to find some representatives for 5 h of market price changes and for each 11×48 feature vector, assign an appropriate representative to it. To this end, in the first step, for each time t of the market, a window of size 5 ($[t+1, \dots, t+5]$) is considered and for each candle in

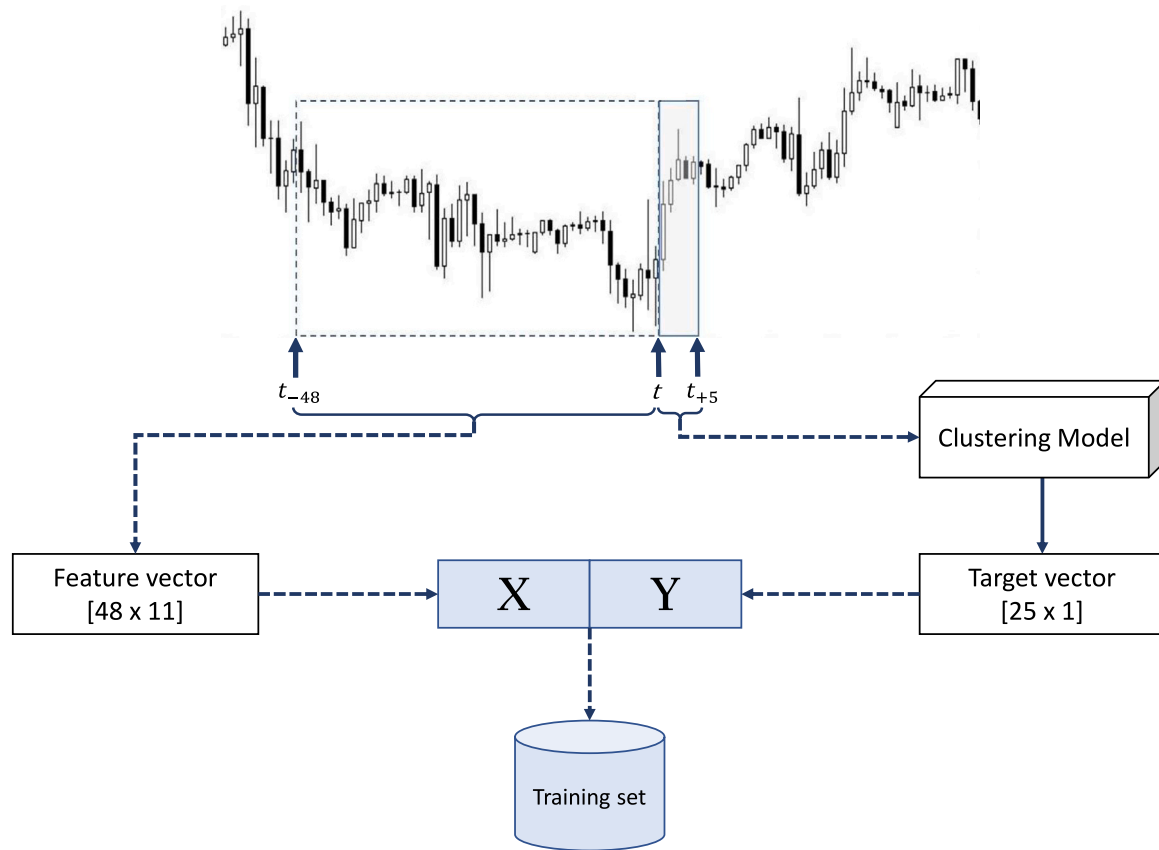


Fig. 8. An illustration of how to extract feature vector for each time t of the market and assign appropriate representative by using clustering model.

this window, the average closing and opening value is calculated. After that, the resultant 5D vectors were used through the k-means clustering algorithm ($k = 25$) to train the clustering model which would be used for representative assignment. By this way, 25 representatives for next 5 h market changes is obtained and utilized to map the market price at each time interval $[t_i, \dots, t_{i+5}]$ to one of the 25 representatives. In other words, the price variation trend in every 5 h is mapped to one of the 25 representatives. In comparison with statistical and regression classification, the clustering algorithm was used because it would generally be impossible to consider a predefined algorithm for every 5-fold vector (the market price in 5 h) to assign a label to the input vector. In fact, there is no available information on the interplay of a vector's features and the clusters patterns. Therefore, an unsupervised algorithm (k-means) was used in this section to assign every 5D vector to one of the 25 possible representatives ($k = 25$). In Section 3.4, we present how to map each representative to the best action (buy, sell and no action). Fig. 8 shows how to create a feature vector for each time t of the market and use the clustering model to assign a representative to the extracted feature vector.

Step 2: Training a deep network model

Deep networks are very efficient in extracting effective features. Using convolutional layers, some of these networks integrate the input vector's features and yield new useful features. In this work, deep convolutional networks were employed and trained by using the training set created in step 1 to adopt the best action. Fig. 9 indicates the proposed structure for the deep network model.

3. Implementation details of the proposed method

As mentioned in Section 2, the general structure of the proposed method consists of two main units (outlier detection unit and best action selection unit) in a consecutive order. Furthermore, these two

Table 3

The number of records used for each symbol.

Symbol	Number of records	Interval
EUR/USD	62 019	2009/1/1–2019/1/1
GBP/USD	61 832	2009/1/1–2019/1/1
USD/JPY	61 837	2009/1/1–2019/1/1

units benefit from totally separate processes of training and parameters optimization. The trained models will finally be placed next to each other. This section addresses the datasets used for training these units first. After that, evaluation criteria are introduced for each unit. Finally, the method of adopting the best action for each representative and the most appropriate anomaly threshold are discussed.

3.1. Dataset

In this study, RoboForex data were employed to train, optimize, and test the proposed models. All of the selected data were from the H1 (one hour) time frame. Table 3 shows the properties and intervals used for each symbol. The models of the proposed method were separately trained for all of the symbols. Each symbol's data were divided into three categories to train, optimize (find the most optimal action for each representative and find the anomaly threshold) and test the specific models of each symbol (70% for training, 15% for optimization, validation and parameters tuning, and 15% for testing).

3.2. Evaluation criterion

In the proposed approach, two models with different structures were integrated to finally develop the proposed system. Therefore, different evaluation criteria were considered for each model and the

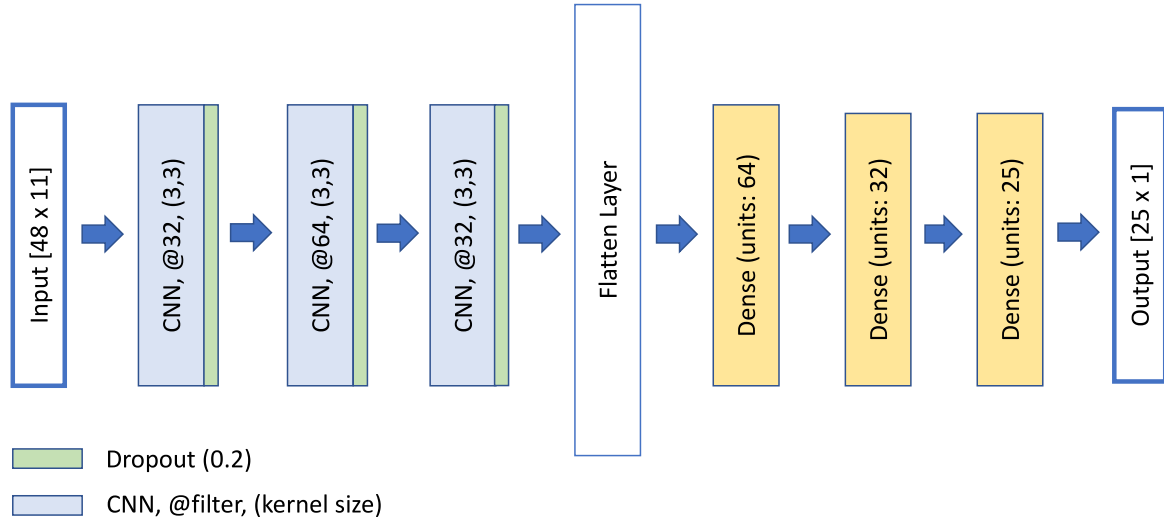


Fig. 9. An illustration of the best action selection network model.

final integrated system. These criteria and the reasons for adopting them are now discussed separately.

3.2.1. Outlier detection model evaluation criterion

According to Section 2.1, an autoencoder network model was used for outlier detection. The output vector of that model resembles its input vector. In other words, the most ideal state of this network occurs when the input vector is exactly as the output vector. Due to the peer-to-peer component comparison of this model, MSE was used as the network evaluation criteria. Eq. (2) shows how to calculate MSE for each data record:

$$MSE = \frac{1}{N} \sum_{i=1}^N (X_i - Y_i)^2 \quad (2)$$

where N refers to the number of components in the input (or output) vector (here, $N = 12$). Moreover, X_i and Y_i indicate the i th components of the input and output vectors, respectively.

3.2.2. The action selection model evaluation criterion

The structure of this model was introduced in Section 2.2; accordingly, this model is doing a classification task. In other words, the model outputs the most probable representative for each input vector. Given the functionality of this model and the fact that the output vector is 25-dimensional one-hot vector (coming from $k = 25$ clusters), cross entropy was used as a criterion for evaluating this network. The numerical value of this criterion is obtained from Eq. (3).

$$CrossEntropy = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C 1_{y_i \in C_c} \log P_{model}[y_i \in C_c] \quad (3)$$

3.2.3. System evaluation criterion

As discussed earlier, the proposed method consists of two outlier detection and action selection models. The decision-making or action selection model was developed by integrating the resulting models to eliminate outliers and select the best action. Total profit and maximum drawdown were used as the evaluation criteria, which are discussed in the following.

Total Profit: Some of the transactions conducted within a specific interval are successful and profitable, whereas the others are unsuccessful and loss-making. Total profit means the subtraction of gross loss from gross profit. This criterion is employed to determine the total profitability of a system. However, this evaluation criterion disregards the risk and chance at which an account might be placed in a loss boundary or be marginalized.

$$TotalProfit = GrossProfit - GrossLoss \quad (4)$$

In Eq. (4), Gross Profit and Gross Loss refer to the net income and net loss of all transactions conducted within a specific interval. The subtraction of the former from the latter equals the total profit within that interval.

Maximum Drawdown: A trader might earn a high profit within a specific interval; however, this evaluation would not be sufficient for the performance assessment because the trader might have put the account on the brink of failure with high-risk transactions and earned a considerable profit with the return of the market price. For the sake of clarity, an example is presented here. With an initial balance of \$1000, a trader opens a buying order for 1 pip and the profit threshold of 10 pips. This means that the trader's balance changes by \$100 per unit (pip) change in market price. If the market falls by 9 pips and then rises by 19 pips, the trader's balance will increase by \$2000 (100% profit). However, the trader might have put 90% of the capital at risk in another transaction, something which is totally illogical and risky with respect to the final profit.

The maximum drawdown was used as a criterion for evaluating the highest risk exposure of the capital. This criterion helps determine the maximum capital decline within a specific interval. Eq. (5) shows how to calculate the maximum drawdown.

$$MaximumDrawdown = \frac{TroughValue - PeakValue}{PeakValue} \quad (5)$$

Within the test period time, the largest difference between the capital balance at t_i and the peak value of the capital balance at t_j , $\forall j < i$ would show the maximum drawdown.

3.3. Optimal outlier threshold

As discussed earlier, the outlier detection unit was employed to find outliers. The anomaly was then quantified. Next, the market simulator was used in the optimization interval to determine the appropriate threshold to either confirm or reject anomalous data. Different symbols of the stock exchange markets have different thresholds. Table 4 shows the thresholds determining data anomaly for a few stock exchange symbols after the optimization phase for a parameter called the *anomaly detection threshold*.

3.4. Adopting the best action for cluster representatives

As discussed in the previous sections, clustering is done on the market 5-h prices to find representative for each time t of the market based on its next 5-h time frame. After that, a deep network model is trained to learn the relation between the 48×11 input vectors

Table 4
Optimal thresholds for outlier detection different symbols.

Symbol	Anomaly detection threshold
EUR/USD	0.2
GBP/USD	0.2105
USD/JPY	19.75

and their corresponding 25×1 cluster representative. These cluster representatives are mapped to one of the possible actions (buy, sell and no action) at last. To find the best action for each representative, total profit is then calculated for each possible action during an optimization time interval and the action that achieves the most total profit is selected as the optimal action for that representative. An important point about the optimal action selection is that it is done for each ticker symbols separately. Table 5 shows optimal actions selected for cluster representatives in EUR/USD symbol.

4. Simulation and analysis results

This section deals with the results of two outlier detection and best action selection units. Finally, the general system performance is assessed through the optimal values found for the system parameters (anomaly detection threshold and allocation of the best action for each cluster representative).

4.1. Results of the outlier detection unit

As discussed in Section 2.1, an autoencoder was employed to detect outlier data characterized by drastic fluctuations in the market. Moreover, the training process was separately performed for each financial market symbol. The model evaluation criterion was discussed in Section 3.2.1. Table 6 presents the results obtained from the models trained within the test period.

According to Table 6, the prices of each symbol have different intervals than other symbols. At the same time, the input data were not normalized in this model in order to prevent outliers from losing their effects on the network training process. Therefore, their real values were used for training a specific network for each symbol. As a result, the accuracy is different for different symbols. For equivalent and normalizing accuracies in different symbols and comparing them with each other, the Symbol Digit parameter can be used through Eq. (6).

$$Accuracy_{normalized} = 10^{Digit_{Symbol}} \times Accuracy \quad (6)$$

4.2. Results of the action selection unit

The action selection unit is responsible for classifying the market data as the labeled clusters ($k = 25$). In other words, this model maps the data records onto one of the clusters representatives. The evaluation criterion for this unit was discussed in Section 3.2.2. Like the outlier detection model, the action selection model was trained separately for each symbol. Table 7 presents the results of models for different symbols within the test period.

Table 7 reports the accuracy obtained before the optimal action was determined for each cluster. In other words, the output vectors of the above models show the cluster id for each data record.

If the distribution of action selection (sell, buy, or no action) is assumed to be equal among clusters with a model selecting a cluster (an action) at random, the probability of a correct selection will be 33.33%. However, the trained models of all symbols presented higher accuracy. For more emphasis, it should be mentioned that selecting a wrong action would cause loss of capitals in addition to bringing no profit. Such an action would also neutralize the previous correct actions (In the simulation process, the profit margin was defined as twice the loss margin. In other words, each wrong action would destroy half of a previous correct action). Thus, the effect of each wrong decision is 1.5 pips.

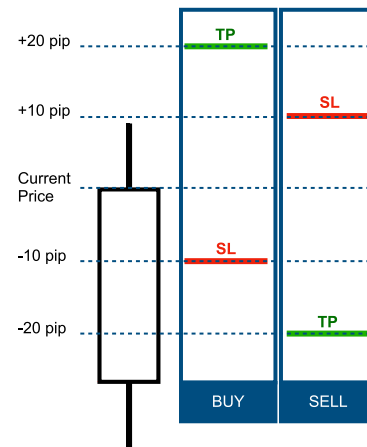


Fig. 10. Take profit (TP) and stop loss (SL) for buy and sell actions.

4.3. Final integrated system results

The proposed approach consists of the units that will lead to an integrated system after the optimal parameters of each unit are obtained and trained for each symbol. The general system evaluation criteria were discussed in Section 3.2.3.

Two strategies, Fixed and Buy&Hold, were considered for order management in the proposed approach. In the Fixed one, after firing action by Action Selection Unit, an appropriate order with fixed take profit and stop loss will be opened and kept open until the price touches either its take profit or stop loss. In addition, the system will not open a new order unless there is not any opened order with that type. In other words, in the Fixed strategy, the maximum number of orders is two (buy and sell). Fig. 10 shows the take profit and stop loss for buy and sell orders.

The other strategy for order management is Buy&Hold (B&H). In this strategy, in contrast with the Fixed one, orders do not have take profit and stop loss. After firing action and opening the appropriate order (buy or sell), it will be kept open as long as Action Selection Unit proposes an opposite action of the order which has already been opened (sell or buy). In this situation, the system will close the elder order and open the new one, immediately.

The performance of proposed approach is evaluated by retraining and testing the system with different configurations (order management and outlier elimination) in different symbols and time frames. Table 8 reports the results of these evaluations.

According to Table 8, the proposed approach managed to yield a profitable performance within the test period. By activating the outlier elimination unit, the approach succeeded in controlling the maximum drawdown and increasing the success rate of transactions. The approach was developed and evaluated in different time frames. However, as the results show, the proposed system is not reliable in long time frames, such as daily. For instance, the number of outputs -cluster indexes- which were assigned to fire a sell action was 0 in the test period of the daily time frame in GBP/USD. Therefore, in the Buy&Hold strategy, after opening the first buy order, it will not be closed throughout the rest of the test period.

To clarify the performance of proposed method in different configurations, Fig. 11 illustrates the balance of all set-ups for EUR/USD in hourly time frame. Also, it can be seen from Fig. 12 that the proposed method was profitable in different hours, weekdays and months.

There are some classical approaches for forecasting time series. The proposed method (Buy&Hold strategy with outlier elimination) was compared with two of them in EUR/USD. Fig. 13 shows the difference of balance between these approaches. The first one was based on

Table 5
Optimal action for each cluster representative in EUR/USD (H1).

ID	Rep.*	Action	ID	Rep.*	Action	ID	Rep.*	Action
1		buy	10		no	19		buy
2		sell	11		sell	20		no
3		buy	12		buy	21		no
4		no**	13		no	22		no
5		sell	14		buy	23		no
6		no	15		no	24		sell
7		sell	16		buy	25		sell
8		buy	17		no			
9		no	18		buy			

* Rep. plots representative of each cluster.

** no stands for no action.

Table 6
Accuracy of the autoencoder network model in different symbols.

Symbol	Symbol digit	Accuracy
EUR/USD	5	7.6555e-7
GBP/USD	5	2.3955e-6
USD/JPY	3	0.0117

Table 7
The action selection model accuracy (without assigning actions to clusters).

Symbol	Accuracy (%)
EUR/USD	38.723
GBP/USD	42.092
USD/JPY	39.194

Autoregressive Integrated Moving Average (ARIMA) and using the Box-Jenkins methodology (Box et al., 2015) for finding the parameters of model. In this approach, the considered variable was the close value of candles. If the predicted value is higher than the close value of the previous candle, the system fires a buy action. Otherwise, it fires a sell action. Each order will be kept open until the opposite action is detected. The other approach was based on the decision-tree model boosted by the Catboost algorithm. The feature vector for this classifier consists of close and moving average values, and the window size was 15. The output of this model predicts that the next close value is higher or lower than the previous close value. Similar to the order management in the ARIMA approach, if the model predicts the higher (lower) value for the next close, the system fires a buy (sell) action.

The network optimization and evaluation processes were performed to compare the proposed system with a few state-of-the-arts within the intervals shown by Table 9.

Table 10 reports the results showing that it is possible to obtain higher total profit at the same time as maintaining the maximum drawdown within the test period. The profit margin, loss margin, and initial capital were considered 20 pips, 10 pips, and \$1000, respectively, and the order management used in the evaluation was Fixed.

According to Table 10, the proposed approach yielded higher total profit than the other methods on the silver exchange market within

the test period. Moreover, the proposed method decreased the maximum drawdown when the outlier detection and elimination unit was activated, although the total profit decreased. Nevertheless, it still yielded more profit than the total profit levels of other methods. In the proposed approach, the profit and loss margins are adjusted in proportion to a trader's initial capital for each transaction. Hence, the loss of each transaction can be controlled. According to the Table 10, the profit and loss margins were considered 20 pips and 10 pips for each deal in this test. The highest loss in each transaction was \$5.29 (assuming the trading lot size to be 0.1).

5. Conclusion

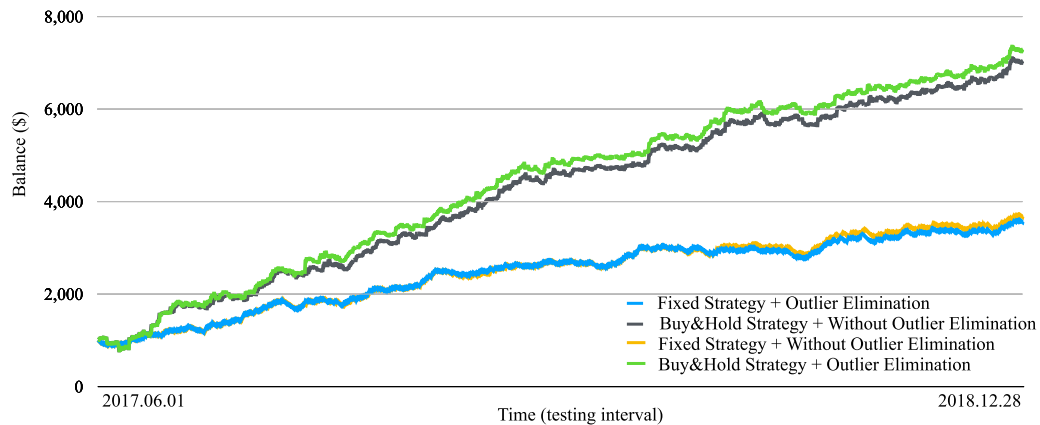
Trading in the financial and stock exchange markets, e.g. the forex market, requires high technical trading knowledge and sufficient time for monitoring the market to find an appropriate opening. Machine learning approaches were used in this study to develop an automated method in which a software robot tracks the market 24 h a day in order to adopt the best action for increasing profit and decreasing drawdown. In this approach, an autoencoder network was used for outlier detection to avoid trading at the time of drastic fluctuations in the market price. When relative stability is detected in the market price trend, the deep-learned model is employed to adopt the best action (sell, buy, or no action) and conduct automated transactions in the system. The trade-orders that opened and managed in this way can be applied to different symbols in the market. For the evaluation of the proposed approach, the system was assessed using a few well-known forex symbols. The results showed the appropriate performance of the proposed system in various symbols.

Nevertheless, the system performance can improve if a few changes are made to the decision-making network structure. One of the main parts of the proposed method is skipping the outlier parts of the market in trading; This is because the following price prediction after those intensive fluctuations might not be accurate. Although the proposed model will be more reliable in this approach, dramatic drops and increases that can be profitable are not involved in trades. Therefore, we aimed to consider outlier patterns in the trading process in the separate line includes specified action selection model. By training and using a model which can predict the future trends after outliers, the

Table 8

Results of the proposed method for different symbols.

Symbol	Time frame	Order management	Outlier elimination	Maximum drawdown	Total profit (\$)	Number of successful trades		Holding position time (HH:MM:SS)		
						Sell	Buy	Min	Max	Avg
EUR/USD	H1	Fixed	No	247.33 (8.03%)	2686.85	981 (37.41%)	1059 (40.23%)	0:00:03	40:18:29	2:04:35
			Yes	283.20 (9.30%)	2574.22	963 (36.97%)	1047 (40.40%)	0:00:15	40:18:29	2:05:47
		B&H	No	280.74 (26.54%)	6037.57	460 (53.26%)	460 (55.22%)	0:55:00	123:00:00	10:42:39
			Yes	280.74 (26.54%)	6287.19	455 (53.63%)	455 (55.38%)	0:55:00	123:00:00	10:49:42
	H4	Fixed	No	330.72 (26.31%)	235.64	607 (33.94%)	268 (37.69%)	0:00:27	26:38:38	2:50:53
			Yes	199.18 (14.45%)	415.51	479 (34.03%)	215 (40.93%)	0:00:27	17:49:35	2:57:28
		B&H	No	637.64 (19.10%)	2365.83	184 (56.52%)	184 (56.52%)	3:54:30	192:00:00	26:48:14
			Yes	637.64 (18.47%)	2480.23	148 (56.08%)	148 (60.14%)	3:54:59	1748:00:00	33:19:26
	Daily	Fixed	No	115.46 (7.31%)	592.21	19 (21.05%)	143 (49.65%)	0:00:59	15:25:27	5:36:27
			Yes	114.28 (7.58%)	521.89	15 (20.00%)	119 (50.42%)	0:00:59	15:25:27	5:38:34
		B&H	No	451.66 (31.73%)	1167.93	14 (64.29%)	14 (71.43%)	23:53:59	1943:59:30	342:51:12
			Yes	1362.99 (102.33%)	-1031.03	9 (22.22%)	10 (60.00%)	23:59:30	2160:00:30	386:31:50
GBP/USD	H1	Fixed	No	117.60 (6.94%)	1028.27	140 (44.29%)	129 (50.39%)	0:00:01	15:34:32	1:29:13
			Yes	118.24 (6.75%)	1055.79	129 (44.19%)	114 (54.39%)	0:00:01	15:34:32	1:33:58
		B&H	No	356.02 (14.05%)	7810.04	51 (68.63%)	50 (68.00%)	1:00:00	649:00:00	94:49:53
			Yes	507.09 (20.64%)	7930.70	48 (72.92%)	47 (68.09%)	1:00:00	649:00:00	100:49:15
	H4	Fixed	No	251.67 (18.84%)	254.68	216 (38.89%)	71 (32.39%)	0:00:19	7:02:15	1:32:43
			Yes	95.39 (8.35%)	243.05	146 (40.41%)	43 (32.56%)	0:01:16	6:44:27	1:36:46
		B&H	No	735.53 (40.94%)	4034.25	47 (59.57%)	47 (59.57%)	4:00:00	1168:00:00	102:48:29
			Yes	513.77 (12.65%)	2880.06	28 (46.43%)	28 (71.43%)	4:00:00	1168:00:00	172:34:16
	Daily	Fixed	No	21.42 (1.90%)	104.33	0 (0.00%)	13 (61.54%)	0:14:32	11:10:54	3:24:50
			Yes	11.90 (1.10%)	116.31	0 (0.00%)	9 (77.78%)	0:14:32	11:10:54	3:28:59
		B&H	No	325.27 (32.53%)	-325.27	0 (0.00%)	1 (0.00%)	9527:53:59	9527:53:59	9527:53:59
			Yes	325.27 (32.53%)	-325.27	0 (0.00%)	1 (0.00%)	9527:53:60	9527:53:60	9527:53:60
USD/JPY	H1	Fixed	No	466.54 (18.33%)	1096.66	897 (36.57%)	679 (37.26%)	0:00:15	57:25:12	2:58:18
			Yes	466.54 (18.20%)	1114.91	896 (36.61%)	678 (37.32%)	0:00:15	57:25:12	0:00:30
		B&H	No	507.74 (7.73%)	5327.54	422 (52.61%)	421 (58.19%)	0:55:00	82:00:00	11:31:23
			Yes	507.74 (7.69%)	5363.25	422 (52.61%)	421 (58.43%)	0:55:00	82:00:00	11:31:23
	H4	Fixed	No	195.31 (13.19%)	852.70	321 (39.25%)	253 (40.32%)	0:00:30	51:03:38	3:03:15
			Yes	195.31 (13.11%)	862.21	321 (39.25%)	252 (40.48%)	0:00:30	51:03:38	3:03:32
		B&H	No	406.50 (14.76%)	3285.20	118 (60.17%)	118 (61.02%)	3:54:00	324:00:00	41:11:09
			Yes	406.50 (14.76%)	3285.21	118 (60.17%)	118 (61.02%)	3:54:00	324:00:00	41:11:09
	Daily	Fixed	No	58.38 (5.51%)	227.35	89 (40.45%)	56 (39.29%)	0:00:30	16:11:40	3:14:49
			Yes	49.81 (4.81%)	215.06	74 (41.89%)	49 (38.78%)	0:00:30	16:11:40	3:23:04
		B&H	No	783.13 (40.66%)	545.54	46 (43.48%)	46 (58.70%)	14:05:00	432:00:00	105:39:03
			Yes	609.33 (53.25%)	325.82	27 (48.15%)	27 (55.56%)	24:00:00	576:00:00	179:33:13

**Fig. 11.** Balances of the proposed approach with different configurations during the test period in EUR/USD (H1).**Table 9**

Intervals of data selection on the silver exchange market.

Title	Interval
Optimization	2014/1/1–2014/4/30
Testing	2014/5/1–2014/12/31

performance and profitability of the system could increase. In addition, some other machine learning structures might be more efficient. For instance, reinforcement learning can be used as a proper alternative to the action selection model. Furthermore, considering and detecting a few sensitive points in the market, such as support and cutoff points, can help make more accurate decisions regarding action selection. These points can be detected by using deep network models. Economic and

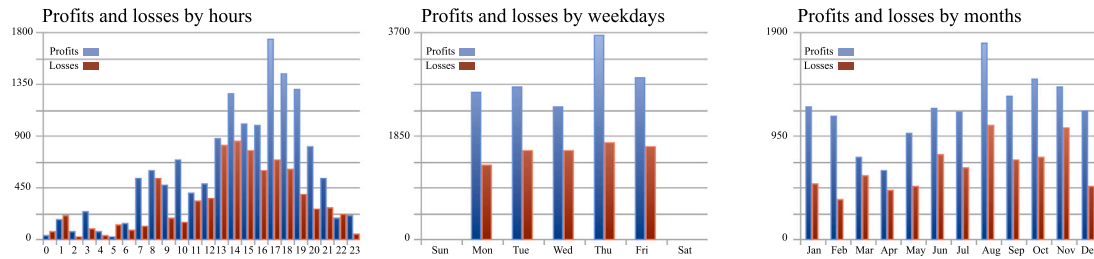


Fig. 12. Profits and losses of the proposed approach (Buy&Hold strategy with outlier elimination) during the test period in EUR/USD (H1).

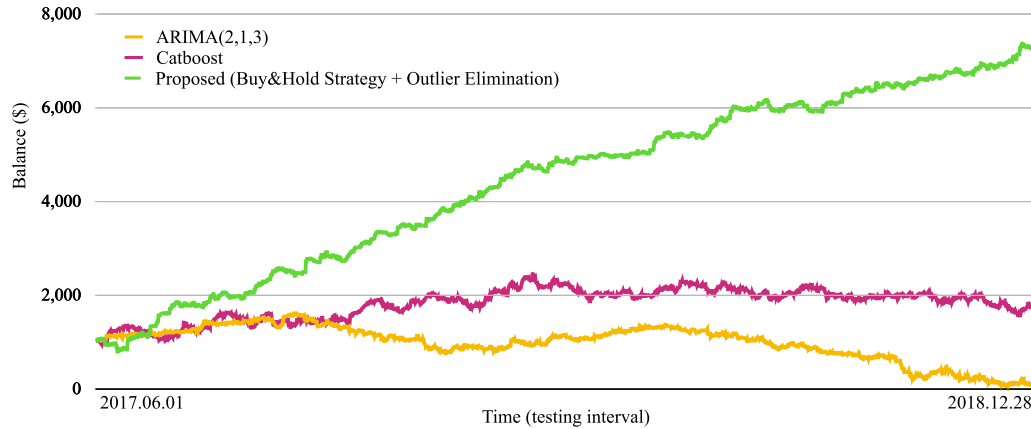


Fig. 13. Comparing balances of classical methods against the proposed approach during the test period in EUR/USD (H1).

Table 10

Comparing the proposed method (with the Fixed order management strategy) against the previous strategies for the silver exchange market.

Approach	Total profit (\$)	Maximum drawdown	Highest risk of each trade (\$)
DRL (Moody & Saffell, 2001)	-113	na ^a	na
SCOT (Deng et al., 2015)	559	na	na
DDR (Deng et al., 2017)	985	na	na
FDDR (Deng et al., 2017)	1567	na	na
Proposed method (without outlier elimination)	1899.05	528.01	5.29
Proposed method (with outlier elimination)	1668.4	422.41	5.29

^ana stands for not available.

political news and controversies of different nations and communities can also affect the trends in the market prices for different symbols. By involving these factors in an NLP-based learning method, it is possible to develop a model that can predict the market trend through news analysis. Thus, we intend to integrate both sides of the market analysis, fundamental and technical approaches, in an ensemble method for future work.

By considering all of the factors discussed to design a more comprehensive system, it is possible to increase instantaneous accuracy for selection of the best action. However, consideration of outlier patterns in a separate flow and using fundamental data, like relevant news, in a parallel-decision line will be considered in the future works of this study.

CRediT authorship contribution statement

Mohammad Reza Roostae: Methodology, Simulation, Writing – original draft. **Ahmad Ali Abin:** Supervision, Conceptualization, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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